

(c) Only 1

(d) Only 2

UPPSC (Pre) 2024

Ans. (c)

'Cereale Yield' is defined as the amount of cereal crops produced per unit area of harvested land. It is generally expressed in kilograms per hectare of harvested land. The cereals measured include wheat, rice, maize, barley, oats, rye millets, sorghum, buckwheat and mixed grains. Therefore, only statement (1) is correct.

101. In the question given below, statement and two assumptions (I) and (II) are given. Choose the correct option from the following :

Statement : In order to replenish the nutrients in the soil, it is important to grow different types of crops in every alternate season.

Assumptions :

(I) A crop can never be grown for the second time in the same field.

(II) If a different crop is grown in the successive season, no additional nutrients such as fertilizers are required to be added to the soil at all.

- (a) Only assumption (I) is implicit.
- (b) Only assumption (II) is implicit.
- (c) Both assumptions (I) and (II) are implicit.
- (d) Neither assumption (I) nor assumption (II) is implicit.

Raj. P.C.S. (Pre) 2024

Ans. (d)

The above statement promotes the periodic change of crops, emphasizing crop diversification (crop rotation) to maintain soil fertility (nutrients). A crop can be grown on the same field again after a few years or within a year interval. Therefore, Assumption (I) is not inherent in the statement. Even when different types of crops are grown in consecutive seasons, fertilizers or manure may still be required to replenish the additional nutrients in the soil. Hence, Assumption (II) is also not inherent in the statement.

102. Which is not an area of Mediterranean type of Agriculture?

- (a) California
- (b) Central Chile

(c) Northern Australia

(d) Southern part of South Africa

Raj. P.C.S. (Pre) 2024

Ans. (c)

The Mediterranean climate is found between 30°-40°/45° latitudes on both sides of the equator, primarily on the western parts of continents. This climatic region is found around the Mediterranean Sea, including Spain, Portugal, France, Southern parts of Italy, Greece, western Turkey, Syria, western Israel and northern Africa. It is also present in California (North America), central Chile (South America), southwestern and southern Australia and the south Western Part of South Africa.

In these regions, Mediterranean type agriculture is practiced, which primarily includes the cultivation of juicy fruits and grapes.

The Minerals

(i) Coal

*When small pieces of rock collect below the surface, it leads to the formation of sedimentary rock also known as layered rocks. Fossils and fossil fuels are found in the sedimentary rocks. Sedimentary rocks occupy 75% of the total Earth's surface. *Minerals such as coal, slate, marble, salt (rock), petroleum are found in sedimentary rocks. Coal and limestone are examples of sedimentary rocks of biotic origin. *Based on the percentage of carbon present coal is categorized into four types—**(1) Anthracite** - It is the best quality of coal having a carbon content of 90%–95%. **(2) Bituminous** – It is next to Anthracite in terms of quality having a carbon content of 70% to 90%. **(3) Lignite** - Famously known as brown coal having a carbon content of 45%–50%. It is mostly used in thermal power plants. **(4) Peat** –It is low-grade coal having a carbon content of 40% or less. **Rock coal** is also known as Anthracite. It is the highest quality of coal. According to the Indian Mineral Yearbook 2022, world proved coal reserves were estimated at 1074108 billion tonnes at the end of 2020 of which 753.639 billion tonnes (70%) have been classified as anthracite & bituminous coal and 320.469 billion tonnes (30%) as sub-bituminous coals & lignite. USA has the largest coal reserves with about 23% share of the total world reserves, followed by Russia (15%), Australia (14%) and China (13%). *According to data from World Mineral Production 2018-2022 in the year 2022, China produced more than half of the world's production of coal. The top five leading producers (2022) of coal in the world are – (1) China, (2) India, (3) Indonesia, (4) USA, (5) Australia. The top 5